

Name: _____
 Class: 12MTZ1
 Teacher: MR WOO

CHERRYBROOK TECHNOLOGY HIGH SCHOOL



2014 AP4

YEAR 12 TRIAL HSC EXAMINATION

MATHEMATICS EXTENSION 2

Time allowed - 3 HOURS
 (Plus 5 minutes reading time)

DIRECTIONS TO CANDIDATES:

- Attempt all questions.
- Multiple choice questions are to be answered on the multiple choice sheet provided at the back of the paper.
- Each question in Section 2 is to be commenced in a new booklet clearly marked Question 11, Question 12, etc on the top of the page and must show your name and class.
- All necessary working should be shown in every question in Section 2. Full marks may not be awarded for careless or badly arranged work.
- Board of Studies approved calculators may be used. Standard Integral Tables are provided.
- Write your name and class in the space provided at the top of this question paper.

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10

- 1 Consider the hyperbola with the equation $\frac{x^2}{144} - \frac{y^2}{25} = 1$.

What are the equations of the directrices?

- (A) $y = \pm \frac{25}{13}$ (B) $y = \pm \frac{144}{13}$
 (C) $x = \pm \frac{25}{13}$ (D) $x = \pm \frac{144}{13}$

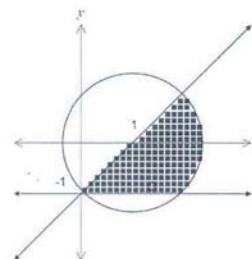
- 2 The points $P(a \cos \theta, b \sin \theta)$ and $Q(a \cos \phi, b \sin \phi)$ lie on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the chord PQ subtends a right angle at $(0,0)$. Which of the following is the correct expression?

- (A) $\tan \theta \tan \phi = -\frac{b^2}{a^2}$ (B) $\tan \theta \tan \phi = -\frac{a^2}{b^2}$
 (C) $\tan \theta \tan \phi = \frac{b^2}{a^2}$ (D) $\tan \theta \tan \phi = \frac{a^2}{b^2}$

- 3 What is $-\sqrt{3} + i$ expressed in modulus-argument form?

- (A) $\sqrt{2}(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6})$ (B) $2(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6})$
 (C) $\sqrt{2}(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6})$ (D) $2(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6})$

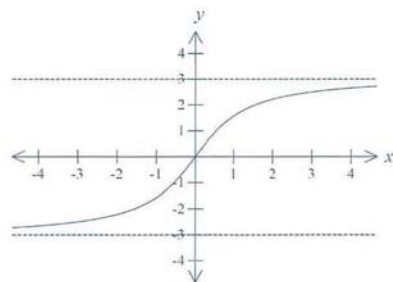
- 4 Consider the Argand diagram below.



Which inequality could define the shaded area?

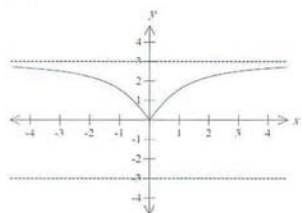
- (A) $|z - 1| \leq \sqrt{2}$ and $0 \leq \arg(z - i) \leq \frac{\pi}{4}$
 (B) $|z - 1| \leq \sqrt{2}$ and $0 \leq \arg(z + i) \leq \frac{\pi}{4}$
 (C) $|z - 1| \leq 1$ and $0 \leq \arg(z - i) \leq \frac{\pi}{4}$
 (D) $|z - 1| \leq 1$ and $0 \leq \arg(z + i) \leq \frac{\pi}{4}$

- 5 The diagram shows the graph of the function $y = f(x)$.

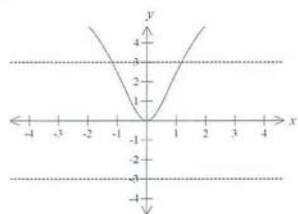


Which of the following is the graph of $y = \sqrt{f(x)}$?

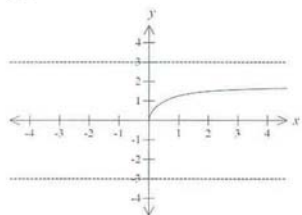
(A)



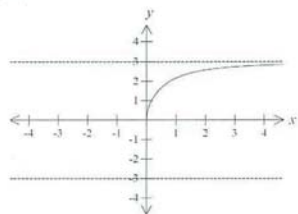
(B)



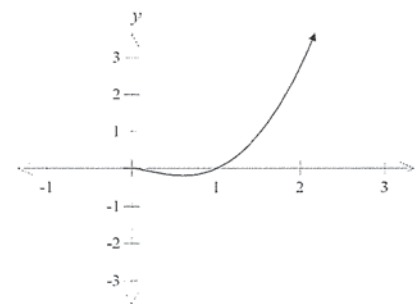
(C)



(D)

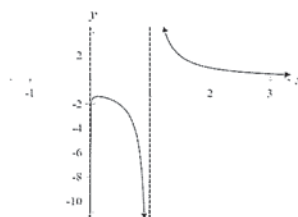


- 6 The diagram shows the graph of the function $y = f(x)$.

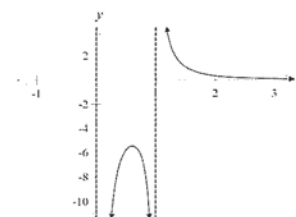


Which of the following is the graph of $y = \frac{1}{f(x)}$?

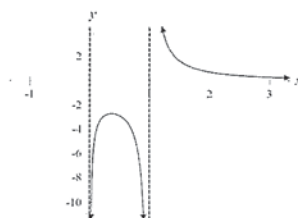
(A)



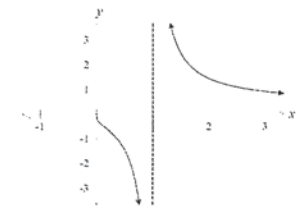
(B)



(C)



(D)



7 Which of the following is an expression for $\int \frac{1}{\sqrt{7-6x-x^2}} dx$?

- (A) $\sin^{-1}\left(\frac{x-3}{2}\right) + c$ (B) $\sin^{-1}\left(\frac{x+3}{2}\right) + c$
(C) $\sin^{-1}\left(\frac{x-3}{4}\right) + c$ (D) $\sin^{-1}\left(\frac{x+3}{4}\right) + c$

8 Which of the following is an expression for $\int \frac{1}{\sqrt{x^2-6x+10}} dx$?

- (A) $\ln\left(x-3-\sqrt{x^2-6x+10}\right) + c$
(B) $\ln\left(x+3-\sqrt{x^2-6x+10}\right) + c$
(C) $\ln\left(x-3+\sqrt{x^2-6x+10}\right) + c$
(D) $\ln\left(x+3+\sqrt{x^2-6x+10}\right) + c$

9 The equation $4x^3 - 27x + k = 0$ has a double root.
What are the possible values of k ?

- (A) ± 4
(B) ± 9
(C) ± 27
(D) $\pm \frac{81}{2}$

10 Given that $(x-1)p(x) = 16x^5 - 20x^3 + 5x - 1$, then if $p(x) = (4x^2 + ax - 1)^2$, the value of a is:

- (A) 1
(B) 2
(C) $\frac{1}{2}$
(D) 0

Section II

90 marks

Attempt Questions 11 – 16

Allow about 2 hours and 45 minutes for this section

Answer each question in a new answer booklet.

All necessary working should be shown in every question.

Question 11 (15 marks) – Use a separate booklet

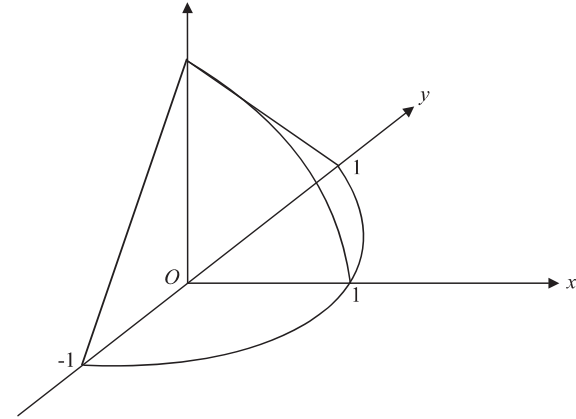
Marks

- (a) Evaluate $\int_0^{\ln 3} \frac{e^x}{1+e^x} dx$. 2
- (b) Use the completion of squares method to find $\int \frac{-2}{\sqrt{3+2x-x^2}} dx$. 2
- (c) Consider the function $f(x) = \frac{x^2-11}{(3x-1)(x+2)^2}$.
- (i) Find real numbers a , b and c such that $f(x) \equiv \frac{a}{3x-1} + \frac{b}{x+2} + \frac{c}{(x+2)^2}$. 3
- (ii) Hence, or otherwise find $\int f(x) dx$. 2
- (d) A point $P(a \sec \theta, b \tan \theta)$ lies on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ with foci $S(ae, 0)$ and $S'(-ae, 0)$.
- (i) Show that $PS = a(e \sec \theta - 1)$ and $PS' = a(e \sec \theta + 1)$. 3
- (ii) Deduce $PS - PS' = 2a$. 3

Question 12 (15 marks) – Use a separate booklet

Marks

- (a) For what values of k does the equation $x^3 - 3x^2 - 24x + k = 0$ have one real root? 4
- (b) Find all the complex numbers $z = a + ib$, where a and b are real, such that $|z|^2 - 7 = 2i(z + 2)$. 4
- (c)



The base of a solid is the semi-circular region in the x - y plane with the straight edge running from the point $(-1, 0)$ to the point $(0, 1)$ and the point $(1, 0)$ on the curved edge of the semicircle.

Each cross-section perpendicular to the x -axis is an isosceles triangle, in which the two equal side lengths are equal to three quarters the length of the third side.

- (i) Show that the area of the triangular cross-section at $x = a$ is given by: 2

$$A(x) = \frac{\sqrt{5}}{2}(1 - a^2)$$

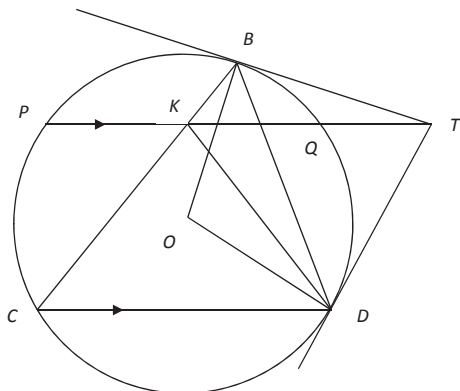
- (ii) Hence find the volume of the solid. 2

- (d) A polynomial $P(x)$ gives remainders -2 and 1 when divided by $2x - 1$ and $x - 2$ respectively. What is the remainder when $P(x)$ is divided by $2x^2 - 5x + 2$? 3

Question 13 (15 marks) – Use a separate booklet

Marks

- (a) The region bounded by the curve $y = \cos x$ and the axes between $x = 0$ and $x = \frac{\pi}{2}$ is rotated about the line $x = 3$ to form a solid of revolution.
- (i) Draw a neat diagram showing the resultant volume. **1**
- (ii) Using the method of cylindrical shells, find the volume of the solid. **4**
- (b)



In the diagram, TB and TD are tangents to the circle (centre O), at B and D . P , K , Q and T are collinear such that $PT \parallel CD$.

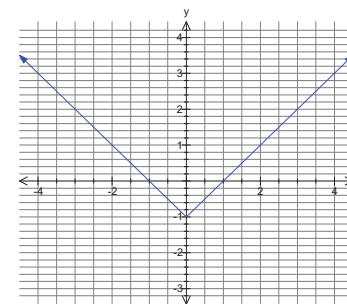
Copy or trace the diagram into your writing booklet.

- (i) Prove that $TBKD$ is a cyclic quadrilateral. **3**
- (ii) Prove that $TBOD$ is a cyclic quadrilateral. **2**
- (iii) Show that K is the midpoint of PQ . **2**
- (c) z is a complex number satisfying the equation $|z + 2 - 4i| = 2$.
- (i) Sketch the locus of z . **2**
- (ii) Find the maximum value of $|z - 1|$. **1**

Question 14 (15 marks) – Use a separate booklet

Marks

- (a) Use the Table of Standard Integrals to find $\int \frac{2x \, dx}{\sqrt{x^4 + 16}}$. **2**
- (b) Consider the graph of the function $f(x) = |x| - 1$ shown below.



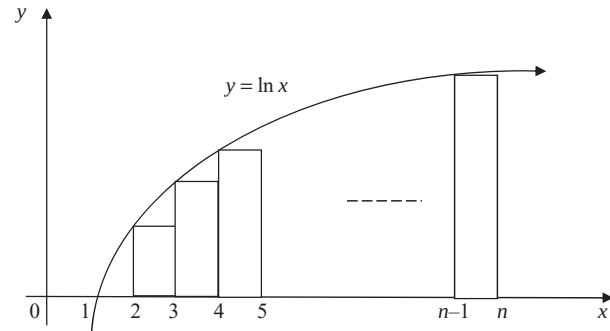
Transformations of $y = f(x)$ are shown below. Draw the resultant graphs on the answer page provided with your multiple choice answer sheet. Show distinguishing features with an accurate scale.

- (i) $y = x f(x)$ **2**
- (ii) $|y| = f(x)$ **2**
- (iii) $y = e^{f(x)}$ **2**

Question 14 continues on the next page

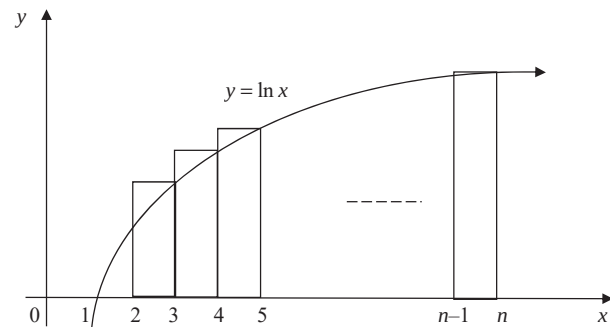
- (c) (i) Evaluate exactly the integral $\int_1^n \ln x \, dx$. 2

- (ii) Consider the curve $y = \ln x$. The area under the curve for $1 \leq x \leq n$ is approximated by dividing it into rectangles under the curve each of width 1 unit. See diagram below (*not to scale*).



Show that the sum of the rectangles, S_u , is given by $S_u = \ln((n-1)!)$.

- (iii) Another approximation, S_o , is made by dividing the area into rectangles that lie above the curve. See diagram below (*not to scale*).



Find a similar expression for this area, S_o .

- (iv) Hence or otherwise show that $(n-1)! < n^n e^{1-n} < n!$. 2

Question 15 (15 marks) – Use a separate booklet

Marks

- (a) Let $z = \cos \frac{\pi}{6} + i \sin \frac{\pi}{6}$. Find z^6 . 2
- (b) (i) Show that the tangent to the rectangular hyperbola $xy = 4$ at the point $P\left(2t, \frac{2}{t}\right)$ has the equation $x + t^2y - 4t = 0$. 2
- (ii) The tangent at P cuts the x axis at point Q . Show that the line through Q which is perpendicular to the tangent at P has the equation $t^2x - y - 4t^3 = 0$. 1
- (iii) The line $t^2x - y - 4t^3 = 0$ cuts the rectangular hyperbola at the points R and S . Show the midpoint M of RS has coordinates $M(2t, -2t^3)$. 2
- (iv) Find the equation of the locus of M as P moves on the rectangular hyperbola. State any restrictions. 2
- (c) The equation $x^4 + bx^3 + cx^2 + dx + e = 0$ has two roots which are reciprocals and two other roots which are opposites.
- (i) Show that $c = 1 + e$. 2
- (ii) Hence, deduce that $d = be$. 2
- (d) Show that $e^x > 1 + x$ for $x > 0$. 2

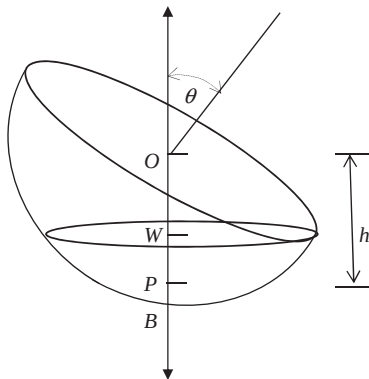
Question 16 (15 marks) – Use a separate booklet

Marks

- (a) **Without** using calculus, draw the graph of: **3**

$$y = \frac{1}{9}(x-1)^2(x^3 - 4x^2 - 15x + 18)$$

 Show all major features of the curve.
- (b) (i) Explain why the cubic $f(x) = x^3 - 3x + 1$ has **exactly** one root, $x = \alpha$, between 0 and 1. **2**
- (ii) Beginning with the approximation $x = 0$, use two applications of Newton's method to show that $\alpha \approx 0.347$ (to 3 decimal places). **2**
- (iii) The diagram below shows a hemispherical bowl of radius r . The bowl has been tilted so that its axis is no longer vertical, but at an angle θ to the vertical. At this angle, it can hold a volume V of water. **2**



The vertical line from the centre O meets the surface of the water at W and meets the bottom of the bowl at B . Let P be any point between W and B and let h be the distance OP .

Explain why $V = \int_{r \sin \theta}^r \pi(r^2 - h^2) dh$.

- (iv) Hence show $V = \frac{r^3 \pi}{3} (2 - 3 \sin \theta + \sin^3 \theta)$ **3**
- (v) The bowl has been tilted so that it is exactly half full. Using the results above, find the angle θ correct to one tenth of a degree. **3**

END OF ASSESSMENT TASK

2014

X2 Trial HSC ~ Multiple Choice Answers

$$1. \quad b^2 = a^2(e^2 - 1) \quad a^2 = 144 \text{ and } b^2 = 25.$$

$$25 = 144(e^2 - 1) \quad a = 12 \quad b = 5$$

$$(e^2 - 1) = \frac{25}{144} \text{ or } e^2 = \frac{169}{144} \text{ or } e = \frac{13}{12}$$

$$\text{Equation of the directrices are } x = \pm \frac{a}{e} = \pm \frac{144}{13}.$$

(D)

$$2. \quad POQ \text{ is a right-angled triangle. Therefore } OP^2 + OQ^2 = PQ^2.$$

$$a^2 \cos^2 \theta + b^2 \sin^2 \theta + a^2 \cos^2 \phi + b^2 \sin^2 \phi = a^2 (\cos \theta - \cos \phi)^2 + b^2 (\sin \theta - \sin \phi)^2$$

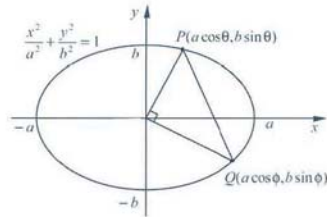
$$a^2 (\cos^2 \theta + \cos^2 \phi) + b^2 (\sin^2 \theta + \sin^2 \phi) = a^2 (\cos \theta - \cos \phi)^2 + b^2 (\sin \theta - \sin \phi)^2$$

Hence

$$0 = -2a^2 \cos \theta \cos \phi - 2b^2 \sin \theta \sin \phi$$

$$2b^2 \sin \theta \sin \phi = -2a^2 \cos \theta \cos \phi$$

$$\frac{\sin \theta \sin \phi}{\cos \theta \cos \phi} = \frac{-2a^2}{2b^2} \text{ or } \tan \theta \tan \phi = -\frac{a^2}{b^2}$$



(B)

$$3. \quad \tan \theta = \frac{1}{-\sqrt{3}}$$

$$\theta = \frac{5\pi}{6}$$

$$r^2 = x^2 + y^2$$

$$= (\sqrt{3})^2 + 1^2$$

$$r = 2$$

$$-\sqrt{3} + i = 2(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6})$$

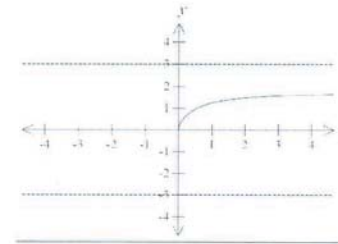
(D)

$$4. \quad |z - 1| \leq \sqrt{2} \text{ represents a region with a centre is } (1, 0) \text{ and radius is less than or equal to } \sqrt{2}.$$

$$0 \leq \arg(z + i) \leq \frac{\pi}{4} \text{ represents a region between angle } 0 \text{ and } \frac{\pi}{4} \text{ whose vertex is } (-1, 0) \text{ not including the vertex } \therefore |z - 1| \leq \sqrt{2} \text{ and } 0 \leq \arg(z + i) \leq \frac{\pi}{4}$$

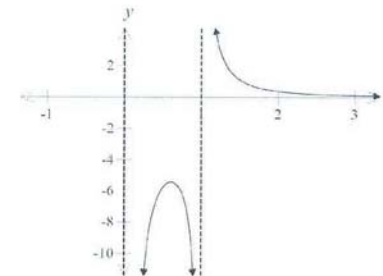
(B)

5.



(C)

6.



(B)

$$7. \quad \int \frac{1}{\sqrt{7 - 6x - x^2}} dx = \int \frac{1}{\sqrt{16 - 9 - 6x - x^2}} dx$$

$$= \int \frac{1}{\sqrt{16 - (x + 3)^2}} dx$$

$$= \sin^{-1} \left(\frac{x + 3}{4} \right) + c$$

(D)

8.

$$\begin{aligned}\int \frac{dx}{\sqrt{x^2 - 6x + 10}} &= \int \frac{dx}{\sqrt{x^2 - 6x + 9 + 1}} = \int \frac{dx}{\sqrt{(x-3)^2 + 1}} \\ &= \ln \left(x - 3 + \sqrt{(x-3)^2 + 1} \right) + c \\ &= \ln \left(x - 3 + \sqrt{x^2 - 6x + 10} \right) + c\end{aligned}$$

9. Let $P(x) = 4x^3 - 27x + k$

$$P'(x) = 12x^2 - 27$$

Let α be the double root.Hence $P(\alpha) = 0$ and $P'(\alpha) = 0$ When $P'(\alpha) = 0$ then $12\alpha^2 - 27 = 0$

$$\begin{aligned}\alpha^2 &= \frac{9}{4} \\ \alpha &= \pm \frac{3}{2}\end{aligned}$$

When $P(\alpha) = 0$ then $4\alpha^3 - 27\alpha + k = 0$

$$\begin{aligned}k &= 27\alpha - 4\alpha^3 \\ &= \alpha(27 - 4\alpha^2) \\ &= \pm \frac{3}{2} \left(27 - 4 \times \frac{9}{4} \right) \\ &= \pm 27\end{aligned}$$

10. $(x-1)(4x^2 + ax - 1)^2 = 16x^5 - 20x^3 + 5x - 1$

Let $x = 2$, $1(15 + 2a)^2 = 16 \cdot 2^5 - 20 \cdot 2^3 + 5 \cdot 2 - 1 = 361$

$$\therefore 15 + 2a = \pm 19$$

$$\therefore 2a = -15 \pm 19 = 4 \text{ or } -34$$

$$\therefore a = 2 \text{ or } -17.$$

(C)

(C)

(B)

Question 11 (a)

Criteria

- One for integration and one for substitution.

Answer:

$$\int_0^{\ln 3} \frac{e^x}{1+e^x} dx = [\ln(1+e^x)]_0^{\ln 3} = \ln(1+e^{\ln 3}) - \ln(1+e^0) = \ln 4 - \ln 2 = \ln 2,$$

Question 11 (b)

Criteria

- One for completing the square and one for simplification.

Answer:

$$\begin{aligned}\text{(b)} \quad \int \frac{-2}{\sqrt{3+2x-x^2}} dx &= -2 \int \frac{1}{\sqrt{-(x^2-2x-3)}} dx \\ &= -2 \int \frac{1}{\sqrt{-(x^2-2x+1-1-3)}} dx \\ &= -2 \int \frac{1}{\sqrt{4-(x-1)^2}} dx \\ &= -2 \sin^{-1} \frac{(x-1)}{2} + c\end{aligned}$$

$$\begin{aligned}u &= x-1 \\ \frac{du}{dx} &= 1\end{aligned}$$

Question 11 (c)

Criteria

- (i) One each for a, b and c. (ii) One for $\ln(x+2)$ and one for simplification

Answer:

(i) By partial fractions

$$\begin{aligned}\frac{x^2-11}{(3x-1)(x+2)^2} &= \frac{a}{(3x-1)} + \frac{b}{(x+2)} + \frac{c}{(x+2)^2} = \frac{a(x+2)^2 + b(3x-1)(x+2) + c(3x-1)}{(3x-1)(x+2)^2} \\ x^2-11 &= a(x+2)^2 + b(3x-1)(x+2) + c(3x-1)\end{aligned}$$

Let $x = -2$

$$(-2)^2 - 11 = c(3(-2) - 1)$$

$$c = 1$$

Let $x = \frac{1}{3}$

$$\left(\frac{1}{3}\right)^2 - 11 = a\left(\frac{1}{3} + 2\right)^2$$

$$\frac{1}{9} - 11 = a \left(\frac{7}{3} \right)^2$$

$$-\frac{98}{9} = \frac{49}{9}a$$

$$-2 = a$$

Let $x = 0$

$$-11 = -2(2)^2 + b(-1)(2) + 1(-1)$$

$$1 = b$$

$$\begin{aligned} \text{(ii)} \quad & \int \frac{x^2 - 11}{(3x-1)(x+2)^2} dx \\ &= \int \frac{-2}{3x-1} + \frac{1}{x+2} + \frac{1}{(x+2)^2} dx \\ &= -\frac{2}{3} \ln(3x-1) + \ln(x+2) + \frac{-1}{x+2} + c \end{aligned}$$

Question 11 (d)

Criteria

- (i) One for substituting into the distance for either PS or PS' and one each for finding the distances PS and PS' (ii) One for explaining $PS - PS' = -2a$, one for explaining $PS - PS' = +2a$ and one for conclusion

Answer:

(a) (i)

$$\text{Length of } PS \text{ is } \sqrt{(a \sec \theta - ae)^2 + (b \tan \theta)^2} = \sqrt{a^2(\sec \theta - e)^2 + b^2 \tan^2 \theta}.$$

$$\text{Given } \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \Rightarrow b^2 = a^2(e^2 - 1).$$

Length of PS :

$$\begin{aligned} &= \sqrt{a^2(\sec \theta - e)^2 + a^2(e^2 - 1)\tan^2 \theta} \\ &= a\sqrt{\sec^2 \theta - 2e \sec \theta + e^2 + e^2 \tan^2 \theta - \tan^2 \theta} \\ &= a\sqrt{e^2(1 + \tan^2 \theta) - 2e \sec \theta + (\sec^2 \theta - \tan^2 \theta)} \\ &= a\sqrt{e^2 \sec^2 \theta - 2e \sec \theta + 1} \\ &= a\sqrt{(e \sec \theta - 1)^2} \\ &= a(e \sec \theta - 1). \end{aligned}$$

Length of PS'

$$\begin{aligned} &= \sqrt{(a \sec \theta + ae)^2 + (b \tan \theta)^2} \\ &= \sqrt{a^2(\sec \theta + e)^2 + b^2 \tan^2 \theta}. \end{aligned}$$

$$\text{Given } \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \text{ and } b^2 = a^2(e^2 - 1).$$

Length PS' :

$$\begin{aligned} &= \sqrt{a^2(\sec \theta + e)^2 + a^2(e^2 - 1)\tan^2 \theta} \\ &= a\sqrt{\sec^2 \theta + 2e \sec \theta + e^2 + e^2 \tan^2 \theta - \tan^2 \theta} \\ &= a\sqrt{e^2(1 + \tan^2 \theta) + 2e \sec \theta + (\sec^2 \theta - \tan^2 \theta)} \\ &= a\sqrt{e^2 \sec^2 \theta + 2e \sec \theta + 1} \\ &= a\sqrt{(e \sec \theta + 1)^2} \end{aligned}$$

Thus the length of PS' is $a(e \sec \theta + 1)$.

(ii) If P lies on the right-hand branch of the hyperbola,

$$\text{then } -\frac{\pi}{2} < \theta < \frac{\pi}{2}.$$

For hyperbola $e > 1$, $PS = a(e \sec \theta - 1)$ and $PS' = a(e \sec \theta + 1)$.

Thus $PS - PS' = -2a$.

If P lies on the left-hand branch of the hyperbola,

$$\text{thus } -\pi < \theta < -\frac{\pi}{2} \text{ or } \frac{\pi}{2} < \theta < \pi.$$

For hyperbola $e > 1$, $PS = -a(e \sec \theta - 1)$ and $PS' = -a(e \sec \theta + 1)$.

Thus $PS - PS' = +2a$. Hence $|PS - PS'| = 2a$.

Question 12(a)

$$\text{Let } P(x) = x^3 - 3x^2 - 24x + k = 0$$

$$P'(x) = 3x^2 - 6x - 24$$

For stationary points ($P'(x) = 0$):

$$3x^2 - 6x - 24 = 0$$

$$x^2 - 2x - 8 = 0$$

$$(x - 4)(x + 2) = 0$$

$$x = 4, x = -2$$

$$\begin{aligned} x = 4, \quad P(4) &= 4^3 - 3(4)^2 - 24(4) + k \\ &= k - 80 \end{aligned}$$

$$\begin{aligned} x = -2, \quad P(-2) &= (-2)^3 - 3(-2)^2 - 24(-2) + k \\ &= k + 28 \end{aligned}$$

$$P(-2) \times P(4) > 0 \text{ (since one real root)}$$

$$(k - 80)(k + 28) > 0$$

$$k < -28, k > 80$$



4 marks	Correct answer
3 marks	Showing $(k - 80)(k + 28) > 0$
2 marks	Showing $P(4) = k - 80$ and $P(-2) = k + 28$
1 mark	Correctly finding x-values of stationary points

Question 12 (b)

$$|z|^2 - 7 = 2i(z + 2)$$

Let $z = a + ib$

$$|z| = \sqrt{a^2 + b^2}$$

$$|z|^2 = a^2 + b^2$$

$$a^2 + b^2 - 7 = 2i(a + ib + 2) \\ = -2b + 2ai + 4i$$

$$a^2 + b^2 + 2b - 2ai = 7 + 4i$$

$$\text{so } -2a = 4$$

$$a = -2$$

$$\text{so } a^2 + b^2 + 2b = 7$$

$$\text{becomes } b^2 + 2b - 3 = 0$$

$$(b + 3)(b - 1) = 0$$

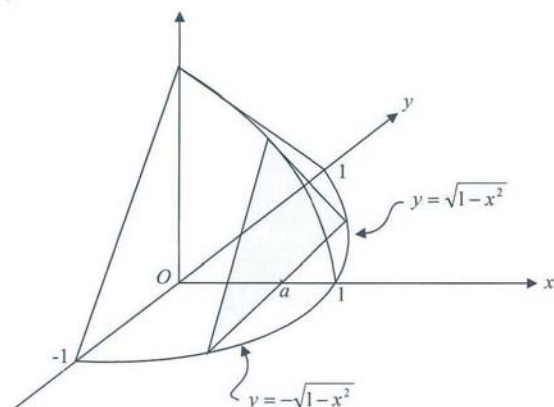
$$b = -3 \text{ or } b = 1$$

$$z = -2 + i, \quad z = -2 - 3i$$

4 marks	Correct answer
3 marks	Obtaining $b = -3$ and $b = 1$
2 marks	Obtaining $a = -2$
1 mark	Obtaining $a^2 + b^2 + 2b - 2ai = 7 + 4i$

Question 12(c)

(b) (i)



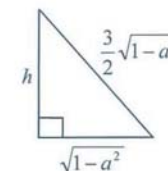
The base of the shaded isosceles triangle has length $2\sqrt{1-a^2}$.

The two equal side lengths are therefore of length

$$\frac{3}{4} \times 2\sqrt{1-a^2} = \frac{3}{2}\sqrt{1-a^2}$$

The height of the isosceles triangle is given by

$$h = \sqrt{\frac{9}{4}(1-a^2) - (1-a^2)} \\ = \sqrt{\frac{5}{4} - \frac{5}{4}a^2} \\ = \sqrt{\frac{5}{4}(1-a^2)}$$



So the area of the isosceles triangle is

$$\frac{1}{2} \times \text{base} \times \text{height} \\ = \frac{1}{2} \times 2\sqrt{1-a^2} \times \sqrt{\frac{5}{4}(1-a^2)} \\ = \sqrt{\frac{5}{4}(1-a^2)^2} \\ = \frac{\sqrt{5}}{2}(1-a^2) \text{ as required.}$$

2 marks	Correct derivation
1 mark	Correct method with one mistake OR identifying the base length of the isosceles triangle as $2\sqrt{1-a^2}$

(ii) Now $\delta V = A \delta x$ where A = area of isosceles triangle

$$= \frac{\sqrt{5}}{2}(1-x^2)\delta x \\ V = \lim_{\delta x \rightarrow 0} \sum_{x=0}^1 \frac{\sqrt{5}}{2}(1-x^2)\delta x \\ = \frac{\sqrt{5}}{2} \int_0^1 (1-x^2) dx \\ = \frac{\sqrt{5}}{2} \left[x - \frac{x^3}{3} \right]_0^1 \\ = \frac{\sqrt{5}}{2} \left\{ \left(1 - \frac{1}{3} \right) - 0 \right\} \\ = \frac{\sqrt{5}}{2} \times \frac{2}{3} \\ = \frac{\sqrt{5}}{3} \text{ cubic units}$$

2 marks	Correct answer
1 mark	Correct method with one mistake OR obtaining $V = \frac{\sqrt{5}}{2} \int_0^1 (1-x^2) dx$

Let $P(x) = (2x^2 - 5x + 2)Q(x) + R(x)$

then $\deg R(x) < 2$

$$P(x) = (2x-1)(x-2)Q(x) + ax + b$$

$$P\left(\frac{1}{2}\right) = \frac{1}{2}a + b = -2 \quad (1)$$

$$P(2) = 2a + b = 1 \quad - (2)$$

$$(2) - (1) \Rightarrow \frac{3a}{2} = 3$$

$$a = 2, b = -3$$

so $R(x) = 2x - 3$

Question 13(a)

$$\delta V = \pi[(3-x)^2 - (3-x-\delta x)^2]y$$

$$= \pi[2(3-x)\delta x - (\delta x)^2]y$$

$$\approx 2\pi(3-x)y\delta x$$

$$V = 2\pi \int_0^{\frac{\pi}{2}} (3-x) \cos x \, dx$$

$$= 2\pi \times [3 \sin x]_0^{\frac{\pi}{2}} - 2\pi \int_0^{\frac{\pi}{2}} x \cos x \, dx$$

$$= 2\pi[3] - 2\pi \left\{ [x \sin x]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \sin x dx \right\}$$

$$= 6\pi - \pi^2 - 2\pi(-1)$$

$$= (8\pi - \pi^2)u^3$$

(i)

$$\angle TDB = \angle BCD$$

(The angle between a tangent and a chord is equal to the angle in the alternate segment.)

$$\angle BCD = \angle BKT$$

(Corresponding angles on parallel lines.)

$$\therefore \angle TDB = \angle BKT$$

TBKD is a cyclic quadrilateral.

(TB subtends equal angles at K and D .)

(ii)

$$OB \perp BT$$

(A tangent is perpendicular to the radius at the point of contact.)

$$\angle OBT = 90^\circ$$

Similarly, $\angle ODT = 90^\circ$

$$\angle OBT + \angle ODT = 180^\circ$$

TBOD is a cyclic quadrilateral.

(If opposite angles of a quadrilateral are supplementary, then the quadrilateral is a cyclic quadrilateral.)

(iii)

Join OK .

From (i) and (ii), $TBKOD$ is a cyclic quadrilateral.

$$\angle OKD = \angle OBD$$

(Angles in the same segment are equal.)

1

$$\angle DKT = \angle DBT$$

(Angles in the same segment are equal.)

$$\begin{aligned}\angle OKD + \angle DKT \\&= \angle OBD + \angle DBT \\&= 90^\circ\end{aligned}$$

(A tangent is perpendicular to the radius at the point of contact.)

$$\therefore OK \perp PQ$$

$$PK = QK$$

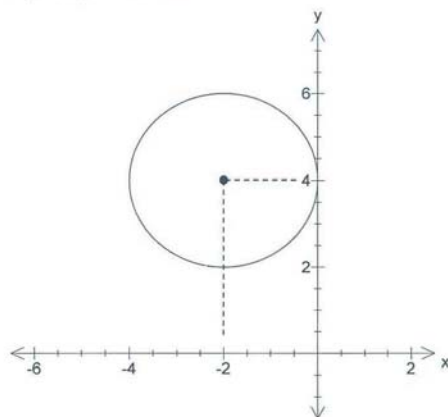
(A perpendicular from the centre to a chord bisects the chord.)

1

Question 13(c)

(i)

Centre = $(-2, 4)$, $r = 2$



2

(ii)

Distance from centre to the point $(1, 0)$

$$\sqrt{(-2 - 1)^2 + (4 - 0)^2} = 5$$

$$\therefore \text{Maximum value } 5 + 2 = 7$$

1

Question 14(a)

$$\int \frac{2x}{\sqrt{x^4 + 16}} dx = \ln(x^2 + \sqrt{x^4 + 16}) + c$$

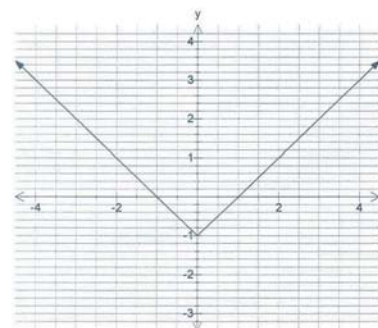
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Question 14(b)

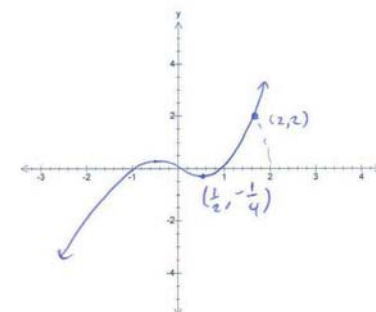
Year 12 Ext 2 Graphing Transformations Answer sheet

DETACH and place in QUESTION 14 booklet

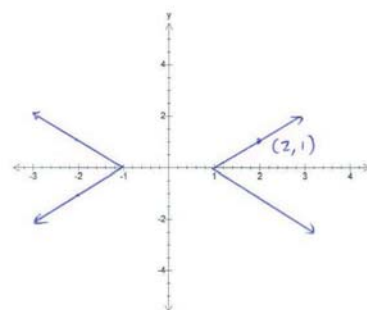
$$y = f(x)$$



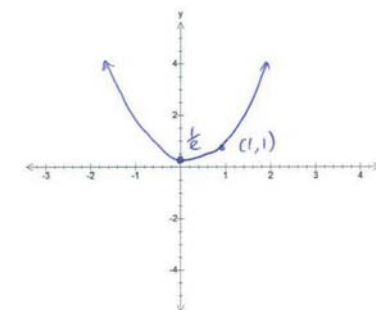
$$y = xf(x)$$



$$|y| = f(x)$$



$$y = e^{f(x)}$$



Question 14(c)

$$\begin{aligned} \text{(i)} \quad \int_1^n \ln x \, dx &= [x \ln x]_1^n - \int_1^n x \cdot \frac{1}{x} \, dx \\ &= n \ln(n) - 0 - [x]_1^n \\ &= n \ln(n) - n + 1 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad S_u &= l_2 b_2 + l_3 b_3 + \dots + l_{n-1} b_{n-1} \\ &= \ln 2 \cdot 1 + \ln 3 \cdot 1 + \dots + \ln(n-1) \cdot 1 \\ &= \ln[2 \cdot 3 \cdot 4 \dots (n-1)] \\ &= \ln[(n-1)!] \end{aligned}$$

$$\text{(iii)} \quad S_a = \ln(n!)$$

$$\begin{aligned} \text{(iv)} \quad S_u &< \int_1^n \ln x \, dx < S_a \\ \ln[(n-1)!] &< n \ln(n) - n + 1 < \ln(n!) \\ \ln[(n-1)!] &< \ln(n^n) - \ln(e^n) + \ln e < \ln(n!) \\ \ln[(n-1)!] &< \ln\left(\frac{n^n \cdot e}{e^n}\right) < \ln(n!) \\ \ln[(n-1)!] &< \ln(n^n \cdot e^{1-n}) < \ln(n!) \\ (n-1)! &< n^n \cdot e^{1-n} < n! \end{aligned}$$

Question 15(a)

$$\begin{aligned} z &= \cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \\ z^6 &= \cos\left(6 \times \frac{\pi}{6}\right) + i \sin\left(6 \times \frac{\pi}{6}\right) \\ &= \cos \pi + i \sin \pi \\ &= -1 \end{aligned}$$

2 Marks: Correct answer.

1 Mark: Uses De Moivre's theorem

Question 15(b)

(i) To find the gradient of the tangent.

$$\begin{aligned} xy &= 4 \\ x \frac{dy}{dx} + y &= 0 \\ \frac{dy}{dx} &= -\frac{y}{x} \\ \text{At } P\left(2t, \frac{2}{t}\right) \end{aligned}$$

$$\begin{aligned} \frac{dy}{dx} &= -\frac{\frac{2}{t}}{2t} \\ &= -\frac{1}{t^2} \end{aligned}$$

Equation of the tangent at P

$$\begin{aligned} y - \frac{2}{t} &= -\frac{1}{t^2}(x - 2t) \\ t^2 y - 2t &= -x + 2t \\ x + t^2 y - 4t &= 0 \end{aligned}$$

2 Marks: Correct answer.

1 Mark: Correctly calculates the gradient of the tangent to the hyperbola

- (ii) Tangent cuts the x axis when $y = 0$.

To find point Q substitute $y = 0$ into $x + t^2y - 4t = 0$.

$$x + t^2 \times 0 - 4t = 0$$

$$x = 4t \quad \therefore Q(4t, 0)$$

At $Q(4t, 0)$ Gradient of the tangent is $-\frac{1}{t^2}$

Gradient of the normal is t^2 ($m_1 m_2 = -1$)

Equation of the normal at Q

$$y - 0 = t^2(x - 4t)$$

$$y = t^2x - 4t^3$$

$$t^2x - y - 4t^3 = 0$$

- (iii) R and S are the points of intersection of normal at Q and the hyperbola.

Solve equations simultaneously

$$xy = 4 \quad (1)$$

$$t^2x - y - 4t^3 = 0 \quad (2)$$

Eqn (1) $\times$ Eqn (2)

$$t^2x^2 - 4t^3x = 4$$

$$t^2x^2 - 4t^3x - 4 = 0$$

$$\text{Sum of the roots is } x_1 + x_2 = -\frac{-4t^3}{t^2} = 4t.$$

(Sum of the roots at R and S)

$$\text{Midpoint of the roots } x = \frac{x_1 + x_2}{2} = \frac{4t}{2} = 2t$$

This is the x coordinate of M .

Substitute $x = 2t$ into the equation of the normal at Q .

$$t^2 \times 2t - y - 4t^3 = 0 \text{ or } y = -2t^3$$

Coordinates of M are $(2t, -2t^3)$

- (iv) Eliminate t from the two parametric equations.

$$x = 2t \quad (1)$$

$$y = -2t^3 \quad (2)$$

Substitute $t = \frac{x}{2}$ from Eqn (1) into Eqn (2).

$$y = -2\left(\frac{x}{2}\right)^3 \text{ or } y = -\frac{x^3}{4}$$

Locus is $x^3 + 4y = 0$ (Excludes case when $t = 0$ (undefined)).

1 Mark: Correct answer.

2 Marks: Correct answer.

1 Mark: Determines $t^2x^2 - 4t^3x - 4 = 0$ or makes similar progress.

2 Marks: Correct answer.

1 Mark: Finds the equation of the locus or states the correct restriction.

Question 15(c)

- (i) Let the roots of $\alpha, \frac{1}{\alpha}, \beta, -\beta$.

$$\sum \alpha\beta\gamma\delta = e$$

$$\alpha \cdot \frac{1}{\alpha} \cdot \beta \cdot (-\beta) = e$$

$$-\beta^2 = e \quad (1)$$

$$\sum \alpha\beta = c$$

$$\alpha \cdot \frac{1}{\alpha} + \alpha\beta + \alpha(-\beta) + \frac{\beta}{\alpha} + \frac{(-\beta)}{\alpha} = c$$

$$1 - \beta^2 = c \quad (2)$$

Solving Eqns (1) and (2) simultaneously.

$$1 + e = c$$

$$c = 1 + e$$

- (ii) $\sum \alpha\beta\gamma = d$

$$\alpha \cdot \frac{1}{\alpha} \cdot \beta + \alpha \cdot \frac{1}{\alpha} \cdot (-\beta) + \frac{1}{\alpha} \cdot \beta \cdot (-\beta) + \alpha \cdot \beta \cdot (-\beta) = d$$

$$-\beta^2\left(\frac{1}{\alpha} + \alpha\right) = d \quad (3)$$

$$\sum \alpha = e$$

$$\alpha + \frac{1}{\alpha} + \beta + (-\beta) = -b$$

$$\alpha + \frac{1}{\alpha} = -b \quad (4)$$

Solving Eqns (3), (4) and (1) simultaneously.

$$eb = d$$

$$d = be$$

2 Marks: Correct answer

1 Mark: Correctly finds relations between roots and coefficients.

2 Marks: Correct answer

1 Mark: Correctly finds relations between roots and coefficients.

Question 15(d)

$$e^x > 1 + x$$

$$\text{or } e^x - 1 - x > 0$$

$$\text{Let } f(x) = e^x - 1 - x$$

$$f'(x) = e^x - 1$$

$$> 0 \text{ when } x > 0$$

Hence $f(x)$ is an increasing function for $x > 0$

Absolute minimum for $f(x)$ is 0 when $x = 0$

Therefore $f(x) > 0$

$$e^x - 1 - x > 0$$

$$e^x > 1 + x$$

2 Marks: Correct answer.

1 Mark: Shows some understanding.

Question 16(a)

$$\text{Let } P(x) = x^3 - 4x^2 - 15x + 18$$

$$P(1) = 1 - 4 - 15 + 18$$

$$= 0$$

$\therefore x - 1$ is a factor of $P(x)$.

$$1 \overline{) 1 \quad -4 \quad -15 \quad 18}$$

$$\underline{1 \quad -3 \quad -18}$$

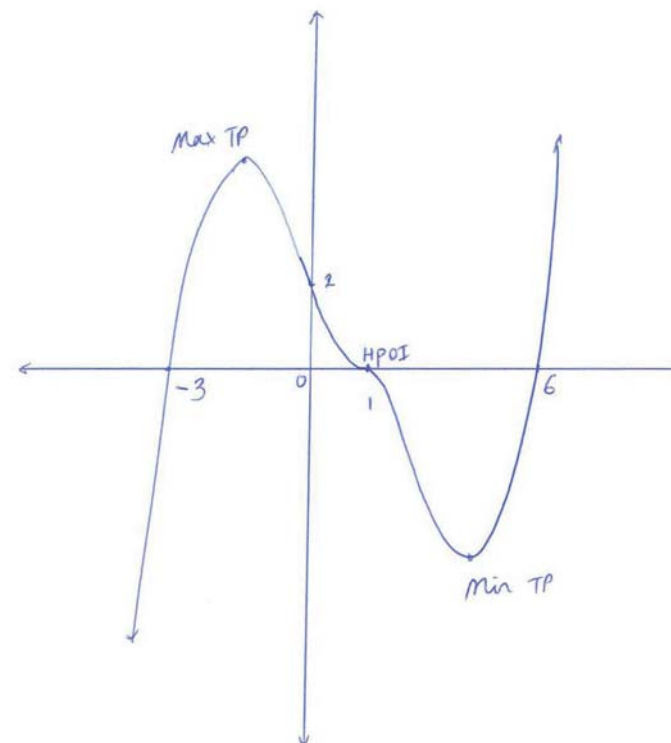
$$1 \quad -3 \quad -18 \quad 0$$

$$x^2 - 3x - 18$$

$$\therefore P(x) = (x - 1)(x^2 - 3x - 18)$$

$$= (x - 1)(x + 3)(x - 6)$$

$$\therefore y = \frac{1}{9}(x - 1)^3(x + 3)(x - 6)$$



Question 16(b)

(i) $f(x) = x^3 - 3x + 1$, $f(0) = 1$, $f(1) = -1$

Hence, there is at least one root between 0 and 1. Note that $f'(x) = 3x^2 - 3$ and $f'(x) = 0$ when $x = \pm 1$ and $f''(x) < 0$ (i.e. monotonic decreasing) in the interval $0 < x < 1$. Hence, there is exactly one root between 0 and 1.

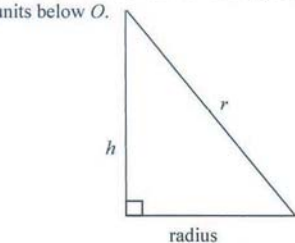
(ii) Let $x_0 = 0$. Using Newton's method,

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = \frac{1}{3} \text{ and}$$

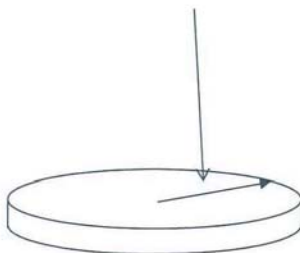
$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = \frac{1}{3} - \frac{\left(\frac{1}{3}\right)^3 - 3 \times \left(\frac{1}{3}\right) + 1}{3 \times \left(\frac{1}{3}\right)^2 - 3} \approx 0.347$$

(iii)

Consider the volume of the water by considering horizontal discs of thickness δh . Consider the disc h units below O .



$$\text{radius}^2 = r^2 - h^2$$



Let δV = Volume of the typical disc

$$= \pi(r^2 - h^2) \delta h$$

The total volume of water $\approx \sum_{h=r \sin \theta}^r \pi(r^2 - h^2) \delta h$

$$= \sum_{h=r \sin \theta}^r \pi(r^2 - h^2) \delta h$$

$$\text{Exact volume} = \lim_{\delta h \rightarrow 0} \sum_{h=r \sin \theta}^r \pi(r^2 - h^2) \delta h$$

$$= \int_{r \sin \theta}^r \pi(r^2 - h^2) dh$$

$$(ii) \quad V = \int_{r \sin \theta}^r \pi(r^2 - h^2) dh$$

$$= \pi \left[r^2 h - \frac{h^3}{3} \right]_{r \sin \theta}^r$$

$$= \pi \left[r^3 - \frac{r^3}{3} - \left(r^3 \sin \theta - \frac{r^3 \sin^3 \theta}{3} \right) \right]$$

$$= \frac{r^3 \pi}{3} (3 - 1 - 3 \sin \theta + \sin^3 \theta)$$

$$= \frac{r^3 \pi}{3} (2 - 3 \sin \theta + \sin^3 \theta)$$

$$(iii) \quad \text{Volume of a hemisphere} = \frac{2}{3} \pi r^3$$

$$\therefore \frac{r^3 \pi}{3} (2 - 3 \sin \theta + \sin^3 \theta) = \frac{1}{3} \pi r^3$$

$$\therefore 2 - 3 \sin \theta + \sin^3 \theta = 1$$

$$\sin^3 \theta - 3 \sin \theta + 1 = 0$$

$$\text{Now, in 14 (c), } \alpha^3 - 3\alpha + 1 = 0 \Rightarrow \alpha \approx 0.347$$

$$\text{Hence, } \sin \theta \approx 0.347 \Rightarrow \theta \approx 20.3^\circ$$

